How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design

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Abstract Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, procedures are followed, and audit trails are complete. This paper introduces the **Regulatory HCI** framework, which reframes regulated environments as interaction systems and establishes a generalizable mechanism by which system design produces accountability conditions under compliance constraints. The framework identifies four interdependent layers at which design choices determine accountability conditions: an accessibility layer, which shapes what information practitioners can reach; a structural layer, which shapes how document relationships are visible and navigable; a coordination layer, which shapes how work is distributed and observable across actors; and a reasoning layer, which shapes how explanations are generated, compared, and stabilized. Grounded in four prototype-based empirical studies conducted in pharmaceutical GMP environments (each examining one layer, each with $N = 20$ GMP-experienced participants), the framework establishes a consistent mechanism across all layers: system design creates conditions that are invisible to formal quality indicators, that redistribute labor unequally across a workforce stratified by experience, and that attribute the resulting burden to individuals rather than to the structural conditions that produced it. The framework extends existing sociotechnical systems theory and value-sensitive design into compliance-constrained environments, offering an account of how responsibility is produced through system structure rather than assigned after the fact.

Keywords: Regulatory HCI, accountability, compliance-constrained design, sociotechnical systems, GMP, judgment preservation, invisible work, interaction design

Introduction

Regulated environments — pharmaceutical manufacturing, clinical healthcare, aviation, financial compliance — are typically evaluated as documentation systems: whether records are accurate, whether procedures are followed, whether audit trails can be reconstructed. This framing treats regulation as a set of requirements imposed on work, and treats system design as the problem of satisfying those requirements efficiently. It asks whether users perform tasks correctly. It does

not ask how system design determines which tasks are available to perform, which interpretations remain open, or which coordination work gets recognized.

I argue that regulated environments should be understood as interaction systems — systems that actively shape how users access information, interpret it, coordinate with others, and ultimately produce accountable decisions. The design of these systems is not neutral with respect to accountability: it determines which explanations remain viable, which pathways are visible, and which forms of labor are measured. Responsibility, in this framing, is not only assigned after the fact. It is produced through the structure of the system itself.

This reframing has practical consequences. If regulated systems are documentation systems, the design goal is accurate record-keeping. If they are interaction systems, the design goal is something different: creating the conditions under which practitioners can reach relevant information, navigate document relationships, observe coordination patterns, and sustain the comparison of alternatives under pressure. These conditions are not captured by any existing quality metric, not measured by deviation reports or CAPA records, and not addressed by compliance-oriented interface design. They constitute an invisible layer of the system — one that shapes accountability outcomes before any formal event occurs.

I introduce the **Regulatory HCI** framework for analyzing this production, and derive three design principles that translate it into actionable guidance. Regulatory HCI examines how system design structures the conditions under which accountability emerges, treating compliance-constrained interaction as a design problem rather than a compliance problem. The framework identifies four interdependent layers at which system design shapes accountability conditions: an accessibility layer, which determines how users reach information; a structural layer, which determines how document relationships are represented and navigable; a coordination layer, which determines how work is distributed and made visible across actors; and a reasoning layer, which determines how explanations are generated, compared, and stabilized (Figure 1).

The framework is grounded in four empirical studies conducted in pharmaceutical GMP environments, each examining one layer through prototype-based evaluation with $N = 20$ GMP-experienced participants. Full accounts of each study are reported in companion papers currently under review: the accessibility layer study (Kim, 2026a), the structural layer study (Kim, 2026b), the coordination layer study (Kim, 2026c), and the reasoning layer study (Kim, 2026d). This paper synthesizes across the four studies to establish the cross-layer mechanism, introduce the design principles that follow from it, and develop the theoretical argument that regulated environments are interaction systems in which responsibility is produced rather than assigned.

This paper makes three contributions. First, it introduces Regulatory HCI as a theoretical framework that reconstitutes regulated environments as interaction systems, shifting the design question from compliance optimization to accountability production. Second, it establishes a

four-layer model linking system design to accountability production across access, structure, coordination, and reasoning. Third, it derives three design principles — upstream stabilization, observability over optimization, and judgment preservation — that translate the framework into actionable guidance for designers working in compliance-constrained environments.

Related Work

Sociotechnical Systems and the Design of Work

The premise of the Regulatory HCI framework — that system design shapes the conditions under which accountability emerges — draws on a long tradition of sociotechnical systems research. Baxter and Sommerville (2011) establish that the design of work systems cannot be separated from the social conditions in which they operate: technical design choices produce organizational outcomes, and organizational conditions shape how technical systems are used. Regulatory HCI extends this argument into compliance-constrained environments specifically — environments in which the sociotechnical stakes of design are particularly high because accountability is formally required and legally consequential. In these environments, the cost of misaligned system design is not merely inefficiency; it is the systematic misattribution of responsibility to individuals whose behavior was structured by conditions the system created and left invisible.

Values in Design

Friedman et al.'s (2002) Value Sensitive Design provides a complementary theoretical foundation. VSD establishes that technology design is never value-neutral: design choices embed assumptions about who matters, what counts as success, and how competing interests are balanced. Regulatory HCI applies this insight to accountability specifically: in regulated environments, design choices embed assumptions about how responsibility is distributed, which forms of labor are recognized, and whose epistemic access to information is treated as a design concern. The four-layer framework operationalizes VSD's core claim by identifying the specific interaction conditions through which these value assumptions are instantiated — and through which they can be redesigned.

Documents as Organizational Action

Harper's (1998) ethnographic account of documentation practices at the IMF establishes the foundational premise of the framework's structural layer: documents are not neutral containers of information but active participants in organizational action. Their accessibility, their linkage

to other documents, and the navigational demands they impose on users are constitutive of organizational practice. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals. In GMP-regulated environments, this observation is amplified by the validated-archive constraint: document structures cannot be modified without regulatory consequence, which means that navigational burden accumulates structurally rather than being addressable through routine system maintenance.

Awareness and the Visibility of Work

Dourish and Bellotti's (1992) account of awareness in collaborative work systems establishes the theoretical basis for the framework's coordination layer. They show that awareness — understanding of the activities of others that provides context for one's own action — is critical to successful coordination and is poorly served by mechanisms that require explicit information-sharing. The framework's coordination layer extends this insight into regulated review workflows: it treats the invisibility of review friction not as a communication failure but as a design failure, and identifies the conditions under which coordination patterns can be made available as ambient, interpretable signal without requiring active compilation by individual participants.

Invisible Work and the Attribution of Labor

Star and Strauss (1999) establish that work performed in the gaps of formal systems tends to become invisible — not captured by indicators, not attributed to the conditions that produced it, and not recognized as labor by the organizations that depend on it. This insight runs through all four layers of the Regulatory HCI framework. At the accessibility layer, retrieval burden is invisible upstream of formal errors: when practitioners consult familiar procedures instead of searching for those most applicable, no record is generated. At the structural layer, traceability reconstruction is invisible in quality metrics: when practitioners manually bridge broken document links, the effort is absorbed individually and unmeasured. At the coordination layer, review friction is invisible in closure records: when delay accumulates through revision cycles, accountability is assigned to outcomes without access to the coordination conditions that produced them. At the reasoning layer, comparison collapse is invisible in final investigation documents: the alternatives that were never meaningfully compared leave no trace in the record that remains. In each case, the system's failure to make conditions visible produces invisible labor that is then attributed to individual failure.

Situated Action and the Conditions of Accountable Practice

Suchman's (1987) account of situated action provides the theoretical spine of the framework. Suchman establishes that action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, accessible, and interpretable at the moment a decision must be made. Regulatory HCI applies this principle systematically across four layers: at each layer, the conditions of action shape the outcomes of action, and those conditions are themselves shaped by system design. The implication is consequential: when accountability is assigned after the fact, based on the final documented outcome, it abstracts away from the conditions under which that outcome was produced. The practitioner whose investigation concluded with a person-centered attribution may have done so not because they reasoned poorly, but because the system did not sustain the conditions under which alternatives could be meaningfully compared. Suchman's framework gives this observation theoretical grounding: situated action is not a deviation from ideal rational procedure, but the normal form of action in all environments.

Human-AI Interaction and the Preservation of Judgment

The framework's reasoning layer draws on Amershi et al.'s (2019) guidelines for human-AI interaction, which establish that AI systems should support human control, make their limitations transparent, and avoid pre-empting human judgment. The framework's judgment preservation principle extends this guideline into a design constraint specific to compliance-constrained environments: in these environments, any computational output that functions as a quality decision introduces a traceability problem for the quality record itself. The goal of AI support is therefore not to accelerate decisions or generate candidate explanations, but to preserve the conditions under which genuine human judgment is possible — keeping alternatives visible, surfacing convergence signals, and resisting institutional pressures toward premature closure.

Epistemic Access and the Distribution of Knowledge

Fricker's (2007) account of epistemic injustice provides theoretical grounding for the framework's unifying theme. Fricker establishes that epistemic access — the ability to know, to be heard as a knower, and to participate in the production of knowledge — is not equally distributed, and that structural conditions determine who can contribute to and benefit from shared knowledge practices. Applied to regulated work systems, design choices which restrict information access, obscure document relationships, or suppress comparison are not merely usability failures: they are conditions that determine whose knowledge counts and who bears the cost of structural invisibility. The experience asymmetry documented across all four empirical studies — in which practitioners with less accumulated implicit knowledge of the system bear

disproportionate labor burden — is an instance of epistemic injustice in Fricker's sense. The system restricts epistemic access not by competence but by failing to externalize the structural knowledge that experienced practitioners have internalized over time.

AI-Assisted Decision Making and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to practitioners navigating complex regulated environments. Wang and Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise, establishing that the design of AI-mediated interaction is not neutral with respect to the judgments it produces. Green and Chen (2019) show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, even when users are instructed to exercise independent evaluation — grounding the design logic of the judgment preservation principle: if interface structure shapes what explanations become salient, redesigning that structure is a tractable intervention against premature convergence. These findings establish that AI support in compliance-constrained environments must be designed with particular care to preserve rather than pre-empt the interpretive authority that regulatory accountability requires.

The Regulatory HCI Framework

Overview

Regulatory HCI is a framework for analyzing how system design shapes accountability conditions in compliance-constrained environments. It identifies four interdependent layers at which design choices determine the conditions under which accountable action is possible: an accessibility layer governing information reach, a structural layer governing document relationship visibility, a coordination layer governing the observability of collaborative work, and a reasoning layer governing how explanations are generated and stabilized (Figure 1).

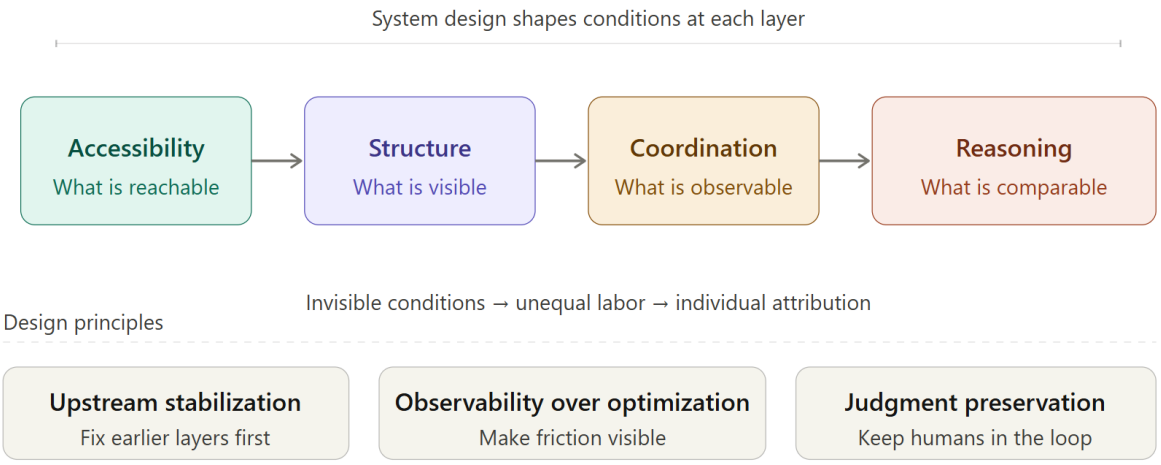


Figure 1: The Regulatory HCI framework: four interdependent layers at which system design shapes accountability conditions in compliance-constrained environments. Arrows indicate the direction of causal dependency: conditions at earlier layers shape the conditions available at later ones.

The framework is grounded in a consistent theoretical commitment: following Suchman (1987), it treats action as situated — shaped by the immediate conditions of its execution rather than by formal plans or procedures alone. This commitment has methodological implications: the framework is not derived from abstract principles about what good system design should look like, but from systematic examination of the interaction conditions that regulated work environments actually produce, and from the gap between those conditions and the accountability outcomes they generate.

The four layers are interdependent in a specific directional sense. Conditions at earlier layers constrain what is possible at later ones: if practitioners cannot reach the relevant procedure (accessibility), they cannot navigate its relationships with other documents (structure), observe the coordination patterns that shaped its use (coordination), or sustain the comparison of explanatory alternatives that would produce accountable investigation conclusions (reasoning). This causal dependency means that interventions at later layers cannot compensate for failures at earlier ones. A system that supports sophisticated reasoning about alternative explanations will not function as intended if practitioners are working from incomplete information because the accessibility layer is poorly designed. Upstream stabilization — the first of the three design principles derived from the framework — follows directly from this structural property.

Layer 1: Accessibility

The accessibility layer concerns how practitioners reach information — the conditions under which documents, procedures, and data are locatable, retrievable, and ready-to-hand at the moment of action. Prior HCI research on professional document work has treated retrieval primarily as a navigational efficiency problem: the goal is to reduce the time and effort required to locate a target document. This framing assumes that the relevant procedure exists and is findable in principle; it asks only how quickly it can be reached. In compliance-constrained environments, this assumption is inadequate.

The question is not only whether a practitioner can locate a procedure, but which procedures become visible enough to be considered, compared, and acted upon at all. When access is constrained, interpretation does not merely slow down — it narrows. A practitioner who cannot locate the most applicable procedure does not suspend action; they act within a personally known subset of the procedural archive, consulting familiar procedures rather than searching for those that might be more directly applicable. In GMP environments, where SOPs define the formal basis for authorized action, this restriction has direct regulatory consequences: the range of procedures a practitioner can justify using is determined not only by what exists in the archive, but by what the archive makes findable under realistic task conditions. The resulting interpretive variability is upstream of any formal error — invisible to deviation records, audit trails, and CAPA analysis alike.

The validated-archive constraint introduces a design boundary with no precedent in general retrieval research: interventions must operate externally, over metadata or retrieval infrastructure, without modifying validated source documents. This constraint is not peripheral — it defines what counts as a permissible intervention in GMP contexts and distinguishes Regulatory HCI from usability-oriented retrieval design. An accessibility intervention in this environment is not a redesign of the archive; it is the construction of an external layer that makes the archive's logic navigable without touching its contents.

Accessibility conditions also distribute unevenly across a workforce stratified by experience. Senior practitioners who have internalized the archive's naming conventions and structural logic can navigate it efficiently; junior practitioners who have not yet built this implicit map bear disproportionate retrieval burden. When the system does not externalize this map, the burden of retrieval falls differentially — producing an accessibility asymmetry that is itself a form of epistemic injustice in the sense that Fricker (2007) describes.

Design implication: accessibility interventions should operate externally, over metadata or retrieval infrastructure, without modifying validated source documents. The design goal is not navigational efficiency alone, but interpretive expansion — making a wider range of procedures reachable, comparable, and defensible at the moment of action, and doing so in a way that reduces rather than amplifies experience-stratified burden.

Layer 2: Structure

The structural layer concerns how document relationships are represented — the conditions under which connections between documents are visible, navigable, and reconstructable. In GMP-regulated environments, documents form interdependent networks: a single deviation investigation may require navigating across standard operating procedures, analytical method protocols, validation records, change control documents, and specification files, each of which references the others. When these reference relationships are complete and current, traceability is possible. When they are missing, outdated, or structurally ambiguous, the burden of reconstructing pathways falls entirely to the practitioner.

The problem is not simply that broken links exist. It is that practitioners cannot distinguish between a broken link and a genuinely absent relationship without manual investigation — and that this investigation labor is performed individually, distributed unequally, and recorded nowhere. A practitioner who manually bridges a broken traceability link performs real work that the system does not measure. If that practitioner is less experienced, they perform more of this work, take longer to complete it, and are more likely to reach an incomplete reconstruction than a senior colleague who has already internalized the implicit map of the document network. The experience asymmetry in loopback behavior — a practitioner's tendency to return to previously visited documents when the path forward is not yet legible — is not a reasoning difference; it is a structural readout of the document system's failure to externalize its own logic.

The validated-archive constraint applies here as well: structural interventions cannot modify validated source files. They must operate externally, making linkage relationships visible and navigable without altering the documents that constitute those relationships. This is a demanding design constraint, but it is also clarifying: the design goal is not to fix the document network, but to make its current state — including its gaps and inconsistencies — legible enough that practitioners can navigate it with shared rather than individually accumulated knowledge.

Design implication: structural interventions should make linkage relationships navigable without modifying validated source files. The goal is to externalize the implicit map of the document network, making it accessible to practitioners regardless of their accumulated experience, and surfacing structural gaps as interpretable information rather than opaque navigational failures.

Layer 3: Coordination

The coordination layer concerns how work is distributed and made visible across actors. In GMP review workflows, deviation records and CAPA documents are reviewed by multiple actors — quality assurance specialists, subject matter experts, department managers — cycling through iterative rounds of comment, revision, and approval. These review processes generate coordination work: comment patterns accumulate, revision cycles compound, and friction

concentrates at specific reviewer roles and document types. This coordination work is real, consequential, and structural — but it remains distributed across individual comment threads, document versions, and reviewer roles, invisible as a pattern to any single participant.

The standard diagnosis of slow review processes — too many reviewers, unclear ownership, insufficient training — misidentifies the problem. The core issue is not delay itself but the invisibility of the coordination work producing it. When coordination patterns are not visible, accountability for outcomes is assigned without access to the conditions that produced them. Delay is attributed to the practitioner or document at the end of the queue, not to the coordination structure that generated it. This attribution is not merely inaccurate; it is structurally produced by a system that does not make coordination conditions available as shared, interpretable signal.

The design challenge is to surface coordination patterns as observable information without automating quality decisions or modifying validated documentation systems. This is the judgment-preservation constraint applied to the coordination layer: the goal is not to generate recommendations about how review should be structured, but to make the current structure of review — including its friction points, asymmetries, and accumulation patterns — visible enough that those with responsibility for it can interpret and act on it with shared rather than partial information.

Design implication: coordination interventions should surface workflow patterns as interpretable signals without modifying validated documentation systems or automating quality decisions. The design goal is observability: making the coordination conditions that produce accountability outcomes visible before those outcomes occur, so that responsibility can be understood in relation to the structural conditions that shaped it.

Layer 4: Reasoning

The reasoning layer concerns how explanations are generated, compared, and stabilized. In regulated investigation workflows — root cause analysis (RCA) in pharmaceutical environments, incident investigation in aviation, adverse event review in clinical healthcare — practitioners are expected to produce causal explanations that are both accurate and accountable. In practice, these investigations unfold under closure pressure: deadlines, documentation requirements, and procedural expectations that define when an investigation must be considered complete. Under these conditions, reasoning does not fail to generate alternatives; it fails to maintain them in a comparable state long enough for systematic evaluation.

The mechanism is not cognitive bias in the individual practitioner's sense. It is structural: the investigation workflow does not sustain alternatives in a comparable state, because the documentation template does not require their explicit comparison, the review process does not reward their retention, and the closure deadline creates pressure to stabilize an account

before comparison is complete. The result is premature accountability convergence — the early stabilization of a causal account that reflects the conditions of its production rather than systematic comparison of alternatives. When this convergence is person-centered, as it typically is in regulated enforcement contexts, the individual practitioner bears the attributed cost of a structural failure.

The design challenge is to preserve comparison as an interaction condition rather than leaving it to individual willpower against institutional pressure. AI support in this layer faces a specific constraint: any system that generates candidate explanations risks narrowing the comparison space by making generated alternatives more salient than unaided ones. The design goal is therefore not explanation generation but comparison enforcement — detecting signals of premature convergence and requiring their reopening, without substituting computational outputs for practitioner judgment.

Design implication: reasoning interventions should preserve comparison between alternatives. AI support in this layer should function as a comparison enforcer rather than an explanation source — detecting convergence signals, surfacing underexplored causal pathways, and requiring explicit evaluation of alternatives before closure is permitted, while leaving the substance of the investigation conclusion to human judgment.

The Mechanism: From Design to Accountability

Across all four layers, a consistent mechanism operates. System design shapes the conditions of action at each layer; the conditions at each layer shape the conditions at the next; and the cumulative effect of design choices across all four layers determines how responsibility is produced and distributed. In each case, design choices that leave conditions invisible, inaccessible, or unsupported redistribute labor and accountability to individuals — while the conditions that produced the redistribution remain invisible in formal records.

This mechanism has a specific form. At each layer, the system creates a condition that is structurally produced but individually absorbed: retrieval burden that practitioners navigate through personal knowledge of the archive; traceability gaps that practitioners bridge through informal investigation; coordination friction that practitioners absorb without shared visibility into its distribution; comparison collapse that practitioners resist or succumb to without structural support for sustained evaluation. In each case, the labor produced by the system's failure to make conditions visible is attributed not to the system but to the practitioner — through quality records, investigation conclusions, and performance assessments that abstract away from the interaction conditions that generated the outcomes they document.

The design implication that follows from this mechanism is not merely that individual interventions at each layer would be beneficial. It is that the four layers form a system, and

that interventions must be understood in terms of their position within it. An accessibility intervention that expands the range of procedures practitioners can reach creates the precondition for structural navigation to be meaningful; a structural intervention that makes document relationships navigable creates the precondition for coordination patterns to be interpretable; a coordination intervention that makes review friction visible creates the precondition for reasoning to proceed with shared rather than partial information about the conditions of deliberation. The framework does not prescribe a single design solution; it identifies the interaction conditions that any adequate solution must address, and the order in which they must be stabilized.

Empirical Program

The Regulatory HCI framework is grounded in four prototype-based studies conducted in pharmaceutical GMP environments. Each study examines one layer of the framework, using a combination of synthetic document corpora, behavioral observation, and pilot evaluation with $N = 20$ GMP-experienced participants. The studies share a common methodological approach: prototype-based evaluation that functions simultaneously as a design intervention and as an analytic probe, making the interaction conditions the framework identifies visible in ways that conventional error analysis cannot (**?**). Participants were recruited from pharmaceutical quality teams with direct experience in GMP documentation and investigation workflows; the same-sample design across studies eliminates between-participant variance in domain knowledge as a confound.

Layer 1: Accessibility (Kim, 2026a)

The accessibility layer study developed and evaluated an external keyword-searchable retrieval layer over a corpus of 500 synthetic GMP SOPs. The prototype was evaluated with $N = 20$ frequent SOP users using a think-aloud protocol designed to surface both navigational behavior and the interpretive conditions under which retrieval decisions were made.

The central finding was not that a retrieval tool reduces search time — though it does, from approximately six minutes to approximately one minute — but that access conditions shape the behavioral repertoire practitioners allow themselves. When SOP retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive rather than the procedures most applicable to the task at hand. This finding extends prior work on SOP non-compliance: the barrier is not motivational but structural, and it operates not only when practitioners fail to retrieve procedures but when the difficulty of retrieval constrains which procedures they consider retrievable. 18 of 20 participants rated the prototype as intuitive; all 20 expressed adoption intent. Participants also described a broader shift: when procedures become

easier to reach, they become easier to compare, easier to justify using, and less dependent on informal colleague mediation.

Layer 2: Structure (Kim, 2026b)

The structural layer study developed and evaluated a graph-based Traceability Explorer over a synthetic corpus of 150 GMP records, and a Cognitive Navigation Explorer that recorded document-to-document transitions, visit frequency, and loopback events during a standardized out-of-specification investigation scenario.

The traceability visibility prototype was associated with a 30–60% reduction in path reconstruction time; all 20 participants completed target pathway reconstruction, rated ease of use at 5.0/5, and expressed adoption intent. The navigation tracking study revealed loopback behavior across all 20 participant sessions, with experienced practitioners ($\geq$5 years) averaging 2–3 loopbacks versus 7–10 for less experienced participants. This experience asymmetry is not a reasoning gap but a design gap: experienced practitioners have internalized a map of the document network that the system itself does not provide. Making linkage structure visible externalizes that map, redistributing epistemic access rather than merely improving navigational efficiency.

Layer 3: Coordination (Kim, 2026c)

The coordination layer study developed and evaluated a six-tab workflow dashboard over a synthetic dataset of 100 deviation-CAPA review records, with a TF-IDF pattern detection layer that identifies recurring issue clusters across review cycles.

All 20 participants rated both dashboard components 5.0/5 on usability and expressed adoption intent. The central finding was that all 20 participants anticipated reduced accountability asymmetry from friction visibility — describing the dashboard not as a tool for managing review more efficiently, but as infrastructure for making coordination asymmetry discussable rather than invisible. This framing reorients the design goal: the dashboard does not eliminate friction; it makes friction legible as an interaction condition that shapes accountability before failure occurs. Participants also identified a dissociation between immediate legibility and final endorsement: the pre-review checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name.

Layer 4: Reasoning (Kim, 2026d)

The reasoning layer study developed and evaluated an AI-supported RCA Reasoning Scaffolder implemented as a Streamlit web application, organizing investigation reasoning into five sequential stages with a convergence detection mechanism that surfaces underexplored causal pathways when individual-blame language is detected.

A two-study empirical program established both the baseline condition and the design response. Study 1 (timed reflection protocol, $N = 20$) showed that 15 of 20 participants produced single-cause, person-centered explanations at the initial timepoint, with confidence ratings averaging 1 out of 5 — indicating that early convergence reflects institutional pressure rather than epistemic certainty. With five additional minutes of unstructured reflection, 18 of 20 broadened their accounts. Study 2 (AI-supported scaffold, same $N = 20$) produced broader causal accounts in 18 of 20 participants; 12 of 15 shifted from person-centered to systemic explanations. All 20 expressed adoption intent while simultaneously identifying the audit trail as a structural barrier to deployment — describing the scaffold as making visible a choice (not to pursue systemic explanation) that was already being made, but without a record.

Cross-Layer Synthesis

Across all four studies, a consistent structural pattern emerges. At each layer, system design creates a condition that is invisible to formal quality indicators. The invisible condition produces labor that is absorbed by individuals, distributed unequally across experience levels, and attributed to individual failure rather than to the structural conditions that produced it. Prototype intervention at each layer demonstrates that making the invisible condition visible — through external retrieval layers, navigable traceability structures, workflow observability dashboards, and comparison-preserving reasoning scaffolds — reduces the labor burden without modifying validated systems or displacing human judgment.

Three cross-layer findings warrant particular attention. First, the experience asymmetry documented across all four studies has a common structural explanation: experienced practitioners have internalized implicit knowledge of the system — its naming conventions, its document linkages, its review patterns, its investigation expectations — that the system itself does not provide. Making conditions visible externalizes this knowledge, redistributing epistemic access rather than merely improving efficiency. Second, the organizational adoption barrier identified across all four studies has a common structural explanation as well: transparency — surfacing conditions that were previously invisible — also makes visible the choices and patterns that those conditions enabled to remain unexamined. Whether organizations can tolerate that legibility is an organizational and regulatory question, not a design one. Third, the dissociation between adoption intent and deployment feasibility, documented most explicitly in the coordination

and reasoning studies, suggests that the limiting factor for Regulatory HCI interventions is not technical feasibility or user acceptance but organizational incentive: the conditions that HCI can make visible are often conditions that regulated organizations have structural reasons not to examine.

Design Principles

The three design principles derived from the Regulatory HCI framework are not prescriptions for specific interface features. They are constraints on the design space — conditions that any adequate intervention in a compliance-constrained environment must satisfy, regardless of the specific layer it addresses. Each principle follows from the empirical program and from the theoretical framework; each names a design goal that existing HCI approaches to regulated environments have not adequately addressed.

Principle 1: Upstream Stabilization

Improving accountability outcomes requires stabilizing conditions at earlier layers before intervening at later ones. This principle follows from the causal dependency structure of the framework: conditions at earlier layers constrain what is possible at later ones, and interventions at later layers cannot compensate for failures at earlier ones.

The practical implication is counterintuitive. In regulated environments, the most visible accountability problems tend to occur at the reasoning layer: investigations conclude with inadequate causal explanations, responsibility is misattributed, CAPA actions fail to address root causes. The natural design response is to intervene at this layer — to provide better investigation templates, richer root cause analysis tools, more structured reasoning scaffolds. The framework suggests this response is incomplete. If practitioners are working from incomplete information because the accessibility layer is poorly designed, the most sophisticated reasoning scaffold will not produce better outcomes; it will produce better-documented accounts of inadequate reasoning. Retrieval system design is not a secondary concern upstream of the real design problem; it is a precondition for structural, coordination, and reasoning interventions to function as intended.

This principle also has implications for how Regulatory HCI interventions should be evaluated. The downstream effects of upstream interventions may not be visible in conventional quality metrics — deviation rates, CAPA closure times, audit findings — because the mechanism operates through conditions that those metrics do not capture. Evaluation strategies for Regulatory HCI interventions need to measure the interaction conditions the framework identifies, not only the downstream outcomes those conditions produce.

Principle 2: Observability over Optimization

The design goal in compliance-constrained environments is not efficiency but visibility. This principle reframes what HCI is for in regulated contexts. Existing approaches to interface design in regulated environments have been dominated by compliance optimization: how can systems be designed so that practitioners more reliably perform the required tasks, complete the required records, and produce the required documentation? Regulatory HCI proposes a different goal: making the conditions of work visible, so that the conditions that shape accountability can be understood, discussed, and redesigned rather than absorbed individually.

Observability over optimization does not mean that efficiency is irrelevant. It means that efficiency interventions that do not increase visibility — that make practitioners faster at performing tasks while leaving the conditions that shape those tasks invisible — are insufficient. The retrieval prototype in the accessibility layer study reduced search time by approximately five minutes; but its more consequential effect was expanding the range of procedures practitioners allowed themselves to consider. The traceability prototype in the structural layer study reduced path reconstruction time by 30–60%; but its more consequential effect was making the document network's structural gaps legible as information rather than opaque navigational failures. Observability is not opposed to efficiency; it is a condition for efficiency interventions to have the effects they are intended to have.

This principle also applies to the design of AI support in compliance-constrained environments. AI systems that optimize for efficiency — generating candidate explanations, recommending actions, accelerating closure — may improve measured performance on formal quality indicators while leaving invisible the conditions that shape that performance. AI systems that optimize for observability — surfacing convergence signals, making comparison conditions visible, flagging patterns of structural friction — create the conditions under which practitioners can exercise genuine judgment rather than efficient compliance.

Principle 3: Judgment Preservation

In regulated environments, any system intervention that automates quality decisions or pre-empts human accountability introduces regulatory risk. Judgment preservation constrains what AI and computational support can do: at each layer, computational support improves the conditions under which human judgment operates without substituting for it.

This principle has a specific technical form. In accessibility interventions, judgment preservation means that retrieval support surfaces candidate documents and explains retrieval basis without recommending which procedure to use. In structural interventions, it means that traceability tools surface linkage gaps and pathway options without determining which pathway is correct. In coordination interventions, it means that workflow dashboards surface friction

patterns and distribution asymmetries without recommending how review should be restructured. In reasoning interventions, it means that AI support detects convergence signals and surfaces alternative pathways without generating causal conclusions. In each case, the computational system improves the conditions of judgment; the substance of judgment remains with the practitioner.

Judgment preservation is not merely a design preference. In GMP-regulated environments, quality decisions must be made and documented by qualified human reviewers; any system that automates or pre-empts those decisions introduces a compliance risk. The regulatory requirement and the HCI principle converge: the appropriate role of computational support in compliance-constrained environments is to expand the conditions under which human judgment can be exercised, not to replace the judgment that compliance requires. This convergence is not coincidental — it reflects a deeper alignment between the regulatory commitment to human accountability and the HCI commitment to preserving the conditions under which meaningful human decision-making is possible.

Discussion

Regulatory HCI as a Reframing

The dominant framing in compliance-oriented interface design treats regulation as a constraint on design. The design problem is to satisfy regulatory requirements efficiently: to create systems that produce accurate records, support procedural compliance, and minimize deviation rates. Within this framing, design is evaluated by its regulatory outcomes, and the practitioner is the variable to be managed — trained, reminded, prompted, and audited toward compliant behavior.

Regulatory HCI inverts this framing. It treats regulation not as a constraint on design but as the context in which accountability is produced, and treats design as a participant in that production. The question is not how to design systems that satisfy regulatory requirements; it is how to design systems that shape accountability conditions in ways that are fair, observable, and supportive of genuine human judgment. Within this framing, design is evaluated by the interaction conditions it creates, not only by the regulatory outcomes those conditions produce. The practitioner is not the variable to be managed but the agent whose judgment the system should support — and whose judgment is currently being shaped by conditions the system creates but does not acknowledge.

This reframing has implications for how regulated organizations understand their own design problems. Many of the conditions the Regulatory HCI framework identifies — inaccessible archives, broken traceability links, invisible review friction, comparison collapse under closure pressure — are familiar to practitioners in regulated environments. What the framework offers

is not a description of new problems but a theoretical account of why existing problems are structurally produced rather than individually caused, and what kinds of design interventions are therefore appropriate. The shift from "why did this practitioner fail to consult the relevant procedure?" to "what conditions did the system create that made the relevant procedure harder to reach than the familiar one?" is a shift in the unit of analysis that has consequences for how regulated organizations invest in design.

Epistemic Access as a Unifying Theme

Across all four layers, the empirical studies converge on a finding the framework makes explicit: system design determines epistemic access — who can see what information, navigate which pathways, observe which patterns, and sustain which comparisons. The practitioners who bear the greatest burden are consistently those with the least accumulated implicit knowledge of the system: less experienced practitioners who have not yet internalized the archive's naming logic; practitioners new to a team who do not yet know the informal map of document relationships; reviewers who do not have access to the coordination patterns that surround the documents they are asked to evaluate; investigators whose reasoning is shaped by closure pressure without access to structural support for sustained comparison.

This asymmetry is not a training gap; it is a design gap. The system distributes epistemic access unequally not because of differences in practitioner competence but because it externalizes the knowledge needed to navigate it for some practitioners (those who have built it through accumulated experience) while requiring others to build it anew through individual effort. Fricker's (2007) account of epistemic injustice is directly applicable: structural conditions restrict epistemic access not by competence but by accumulated implicit knowledge the system itself does not provide. Making conditions visible — externalizing the archive's logic, the document network's structure, the review workflow's patterns, the investigation's comparison space — is an act of redistributive design: it transfers knowledge from the implicit possession of the experienced to the explicit availability of all.

The Limits of Interface Design

The organizational adoption barriers identified across the empirical studies point to a limitation that interface design alone cannot address. In every study, all participants expressed adoption intent; in every study, participants simultaneously identified structural barriers to deployment. The pattern is consistent: practitioners want the conditions the Regulatory HCI tools instantiate — better access, better traceability, better coordination visibility, better comparison support — but the organizations in which they work may not be organized to support or incentivize that transparency.

The barrier is not technical. It is organizational. Making coordination patterns visible creates records of coordination failures that did not previously exist in documented form. Making comparison collapse visible creates records of investigation choices that were previously implicit and unexamined. Whether regulated organizations can tolerate — or are required by their regulatory environment to produce — that level of transparency is a question that Regulatory HCI cannot answer through design. It requires engagement with quality culture, audit standards, institutional accountability structures, and the regulatory frameworks that govern them. Interface design can make conditions visible; it cannot determine whether organizations choose to act on what visibility reveals.

This limitation does not diminish the value of Regulatory HCI as a framework. It specifies the boundary of what design can do and identifies the organizational and regulatory work that must accompany it. The most important implication may be this: Regulatory HCI tools are not supplements to existing compliance systems. They are challenges to the accountability assumptions embedded in those systems — assumptions that invisible labor is individual failure, that investigation conclusions reflect reasoning quality rather than reasoning conditions, that coordination delay is a practitioner problem rather than a system property. Deploying Regulatory HCI tools in live regulated environments will require organizational readiness to confront those assumptions, not merely technical capacity to implement the interfaces.

Generalizability

The framework was developed in pharmaceutical GMP environments, but the mechanism it identifies is not domain-specific. Any regulated environment in which accountability is formally required, system design is consequential, and the conditions of practice are shaped by interaction with document systems, coordination workflows, and investigation procedures is a candidate for Regulatory HCI analysis.

In clinical healthcare, the accessibility layer manifests in how clinicians reach relevant guidelines, protocols, and patient records at the moment of decision; the structural layer in how clinical documentation systems represent relationships between diagnoses, treatments, and outcomes; the coordination layer in how care coordination across providers and settings is made visible; the reasoning layer in how clinical reasoning under time pressure stabilizes on diagnoses before differential evaluation is complete. In aviation maintenance, the accessibility layer manifests in how technicians reach relevant maintenance procedures and airworthiness directives; the structural layer in how aircraft documentation systems represent relationships between components, maintenance histories, and regulatory requirements; the coordination layer in how maintenance workflows are distributed and made visible across shifts and teams; the reasoning layer in how incident investigations stabilize on causal accounts before systemic

factors have been adequately examined. In financial compliance, legal practice, and public administration, analogous layer-level conditions obtain.

The generalizability claim is not that the specific interaction conditions documented in pharmaceutical GMP environments will replicate exactly in other domains. It is that the mechanism — system design creating invisible conditions that redistribute labor and accountability to individuals — is a property of compliance-constrained interaction as such, and that the four-layer framework provides an analytic structure for identifying how that mechanism operates in any specific regulated environment.

Conclusion

This paper introduced the Regulatory HCI framework for explaining how regulated systems produce responsibility through interaction conditions. The four-layer model — accessibility, structure, coordination, and reasoning — identifies a consistent mechanism across compliance-constrained environments: system design creates conditions invisible to formal quality indicators, redistributes labor unequally across experience levels, and attributes the resulting burden to individuals rather than to the structural conditions that produced it.

Three design principles follow from this analysis. Upstream stabilization holds that accountability conditions must be addressed at earlier layers before interventions at later ones can be effective — that retrieval system design is a precondition for investigation quality, not a secondary concern. Observability over optimization holds that the design goal in compliance-constrained environments is visibility rather than efficiency — that making conditions legible to participants is more consequential than making individual tasks faster. Judgment preservation holds that computational support in regulated environments must improve the conditions under which human judgment operates rather than substituting for it — that the design goal is not to automate accountability but to create the interaction conditions under which accountability can be genuinely exercised.

Together, these principles define a design agenda for Regulatory HCI: the development of external observability layers that make the conditions of compliance-constrained practice visible without modifying the validated systems that compliance requires, and without displacing the human judgment that accountability demands. The four empirical studies reported here demonstrate that this agenda is technically feasible and practically valued by the practitioners who bear the costs of its current absence. The organizational and regulatory work required to deploy it remains to be done.

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